

POINT ESTIMATION

[SLIDES 1-10]

CASES	θ PARAMETER	$\hat{\theta}$ ESTIMATOR	$\hat{\theta}$ ESTIMATE	ESTIMATOR DISTRIBUTION	ESTIMATOR STANDARD ERROR
$X_1, X_2, \dots, X_n$ i.i.d. $E[X_i] = \mu$ $\text{Var}(X_i) = \sigma^2$	μ	$\bar{X}$	$\bar{x}$	$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$	$\sigma/\sqrt{n}$
$X_1, X_2, \dots, X_n$ i.i.d. $X_i \sim \text{Bernoulli}(p)$ $E[X_i] = p$ $\text{Var}(X_i) = p(1-p)$	p	$\hat{P}$	$\hat{p}$	$\hat{P} \sim N(p, \frac{p(1-p)}{n})$	$\sqrt{\frac{p(1-p)}{n}}$

or: $n > 30$

We are interested in a population parameter θ (e.g. μ, p). To infer on it we use an estimator $\hat{\theta}$ (e.g. $\bar{X}, \hat{P}$). The estimator $\hat{\theta}$ is used to estimate θ , and it is a r.v. that depends on a random sample. If we compute the value of $\hat{\theta}$ for a specific sample we get an estimate $\hat{\theta}$ (e.g. $\bar{x}, \hat{p}$) of the parameter θ .

PROPERTIES OF ESTIMATORS [find this in slides 1-10]

θ : PARAMETER (μ, p): feature of the population, generally unknown

$\hat{\theta}$: ESTIMATOR ($\bar{X}, \hat{P}$): r.v. which depend on the random sample.

We use $\hat{\theta}$ to make inference on θ

$\hat{\theta}$: ESTIMATE ($\bar{x}, \hat{p}$): realisation of the estimator $\hat{\theta}$ in correspondence to a specific sample

I UNBIASEDNESS

An estimator $\hat{\theta}$ is UNBIASED if $E[\hat{\theta}] = \theta$

e.g. $E[\bar{X}] = \mu$, $E[\hat{P}] = p$ hence both sample mean and sample proportion are unbiased estimators of μ and p respectively.
population mean population proportion

e.g. Recall the sample variance $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

S^2 is an unbiased estimator of σ^2 (result with no proof)

$$E[S^2] = \sigma^2$$

II BIAS

If an estimator $\hat{\theta}$ is biased then $E[\hat{\theta}] \neq \theta$

In this case the BIAS is defined

$$\text{BIAS}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

e.g. $E\left[\frac{1}{n} \sum (X_i - \bar{X})^2\right] = \sigma^2 - \frac{1}{n} \sigma^2 < \sigma^2$ hence this estimator is underestimating σ^2

this is a biased estimator of σ^2 .

$$\text{Its bias is } = -\frac{\sigma^2}{n} := \text{BIAS}\left(\frac{1}{n} \sum (X_i - \bar{X})^2\right)$$

Obv for an unbiased estimator the bias is zero.

III) ASYMPTOTIC UNBIASEDNESS

An estimator $\hat{\theta}$ is ASYMPTOTICALLY UNBIASED if

$$\lim_{n \rightarrow \infty} \text{BIAS}(\hat{\theta}) = 0 \text{ or equivalently } \lim_{n \rightarrow \infty} E[\hat{\theta}] = \theta$$

e.g. $\frac{1}{n} \sum (X_i - \bar{X})^2$ is an asymptotically unbiased estimator for σ^2

IV) EFFICIENCY

An unbiased estimator is more EFFICIENT than another unbiased estimator if it has a lower **STANDARD ERROR**

standard error is the standard deviation of an unbiased estimator

e.g. $\bar{X}_{30} = \frac{1}{30} \sum_{i=1}^{30} X_i$

(vs)

$$\bar{X}_{1000} = \frac{1}{1000} \sum_{i=1}^{1000} X_i$$

• Both $\bar{X}_{30}$ and $\bar{X}_{1000}$ are unbiased for μ .

• $SE(\bar{X}_{30}) = \sigma / \sqrt{30}$

$$SE(\bar{X}_{1000}) = \sigma / \sqrt{1000}$$

more efficient

Covered so far... [READING]

- ▶ Descriptive statistics: description of the characteristics of a set of observations with reference to a specific variable. Observations might consist of a **sample** or of the **entire population**.
 - ▶ Measures of (central or non-central) tendency and dispersion, calculation of the frequency with which values of a variable are observed; calculation of measures summarizing the extent of the relationship between two variables (correlation and covariance)
- ▶ Random variables: description of the outcome of a **random** sampling of a unit from a population with **known** characteristics.
 - ▶ It is not possible to know with certainty what the observed result will be for a randomly drawn unit: a random variable describes the probabilities with which specific values are observed, as well as the characteristics of the population (e.g. expected value and variance)
 - ▶ In the case of repeated **independent** draws from the **same** population, it is not possible to know with certainty what the values of the calculated statistics (e.g. the mean) will be. However, we have seen that for some statistics it is possible to assess aspects of their probability distribution (expected value and variance, probability of observing specific values) and - in some cases (normal distribution or central limit theorem) - their distribution is known.

Inference [READING]

Inferential statistics concerns the procedures required to make **extrapolations** on **parameters** of a population X from **statistics** (e.g. the sample mean) calculated for a simple random sample $X_1, X_2, \dots, X_n$ extracted from X

- ▶ **Estimation** of a parameter
 - ▶ **Point** estimation: a single value
 - ▶ **Interval** estimation: a range of values
- ▶ **Hypothesis testing** for a parameter: assess which of two possible hypotheses is more supported by sample results

Point estimation: parameter, estimator, estimate

With reference to point estimation, it is fundamental to distinguish between:

- ▶ **Parameter: θ .** Measurable characteristic of the **population with reference to a r.v. X**
Examples of parameters are the mean, μ , or the standard deviation, σ , of the population, or the proportion p of cases with a certain characteristic in the population.
- ▶ **Estimator: $\hat{\theta}$.** **Statistic** used to **estimate θ** . $\hat{\theta}$ is a function of the outcome of (simple) random sample of n i. i. d. draws $X_1, \dots, X_n$ distributed as X , i.e. $\hat{\theta} = f(X_1, \dots, X_n)$. Since the realisations of $X_1, \dots, X_n$ are **random**, $\hat{\theta}$ is a **r.v.** describing the realisations of $\hat{\theta}$ with respect to all the samples that can be extracted from X .
For example, the sample mean $\bar{X} = [(X_1 + \dots + X_n)/n]$ is a possible estimator of the population mean, μ . Note that until a sample is actually drawn, $\bar{X}$ is a random variable, whose realisation is not known a priori.
- ▶ **Estimate: $\hat{\vartheta}$.** **Sample realisation of an estimator $\hat{\theta}$** calculated from the specific sample actually drawn, $x_1, \dots, x_n$.
For example, extracting a **specific** sample of size 10 from X , (1, 0.5, 2.5, 4, 3.5, 6, 7, 8, 8.5, 9) the realisation of $\bar{X}$ is $\bar{x} = 5$.

Point estimation: parameter, estimator, estimate

Examples. We are interested in assessing a master's degree with reference to the placement of graduates. A simple random sample of 200 graduates one year after graduation is drawn.

- ▶ **What is the monthly salary a generic graduate can expect?**
 - ▶ X = monthly salary of a randomly selected graduate
 - ▶ Parameter of interest: population average, μ , calculated on all graduates
 - ▶ Estimator: sample mean (random variable)
- ▶ **What is the (squared) expected deviation around the average salary?**
 - ▶ X = monthly salary of a randomly selected graduate
 - ▶ Parameter of interest: population variance, σ^2
 - ▶ Estimator: sample variance (random variable)
- ▶ **What percentage of graduates found a job within 6 months of graduation?**
 - ▶ X = Bernoulli r.v. indicating whether a graduate found (1) or not (0) work within 6 months
 - ▶ Parameter of interest: proportion of graduates who found a job within 6 months, p
 - ▶ Estimator: sample success rate, $\hat{P}$ (random variable, average of $X_1, X_2, \dots, X_{200}$)

Point estimation: parameter, estimator, estimate [READING]

Unlike when we considered extractions from a population described by a r.v. with **known** parameters, in the inferential case the characteristics of the population are **not** known. The inference will necessarily have to be based on the specific estimate corresponding to a specific sample extracted from the population.

However, to assess the reliability of the inferential procedure, we can ask ourselves:

- ▶ What is the **expected value** of the **estimator we decide to use to estimate a certain parameter**? Does it coincide with the parameter of interest or not?
- ▶ What is the dispersion of the **estimates** around the parameter of interest? Can we expect realisations of the estimator far away from the parameter of interest? And with what probability? With what probability will we extract a sample that yields estimates very far from the parameter?

We will assess the reliability of the inferential process by choosing the estimator with the best properties. However, given the estimate calculated from a specific sample, we will not be able to make any assessment, especially with respect to how close (if close at all!) this estimate is to the value of the parameter (which is unknown).

Properties of estimators: unbiasedness

I

A point estimator $\hat{\theta}_n$ based on a (simple random) sample of n units is said to be **unbiased** for a parameter θ if its expected value coincides with the parameter itself, for any value of n and θ :

$$E(\hat{\theta}_n) = \theta \text{ for any value of } n \text{ and } \theta$$

II

If an estimator $\hat{\theta}_n$ is **biased** for a parameter θ , its **bias** is defined by

$$D(\hat{\theta}_n) = [E(\hat{\theta}_n) - \theta]$$

III

A **biased** estimator for a parameter θ , is said to be **asymptotically unbiased** if, as the sample size increases, the bias is reduced:

$$\lim_{n \rightarrow \infty} D(\hat{\theta}_n) = 0 \text{ or equivalently } \lim_{n \rightarrow \infty} E(\hat{\theta}_n) = \theta$$

Standard error of an unbiased estimator

For a generic **unbiased** estimator $\hat{\theta}$ for a parameter θ , i.e. such that $E(\hat{\theta}) = \theta$:

$$E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - \underbrace{E(\hat{\theta})}_{=\theta})^2] = \text{Var}(\hat{\theta})$$

as $\hat{\theta}$ is unbiased

The **variance of an unbiased estimator is the expected squared deviation of the estimator from the parameter of interest** (as the expected value of the estimator is the parameter itself).

The standard deviation of an unbiased estimator is called standard error of the estimator.

The variance (and the standard error) of an unbiased estimator provides information on the **reliability** of the estimator.

Among all the unbiased estimators for a certain parameter, the one with the smallest **variance (or standard error)** will be preferable, as it guarantees the **greatest concentration of estimates around the parameter**.

we want an estimator $\hat{\theta}$ to be **UNBIASED** and with lowest possible **STANDARD ERROR**

Estimator for the population mean

Given a population described by a r.v. X , we are typically interested in its **mean** (or expected value) $E(X)=\mu$. A 'natural' (or intuitive) estimator for μ based on an i.i.d. random sample $X_1, X_2, \dots, X_n$, is the **sample mean** $\bar{X}$.

As we have already seen, **whatever the distribution of X** is, the following hold:

- ▶ the expected value of $\bar{X}$ is: $E(\bar{X}) = [E(X_1) + \dots + E(X_n)]/n = n \cdot \mu / n = \mu$
therefore $\bar{X}$ is an **unbiased estimator for μ**
- ▶ The variance of $\bar{X}$ is: $Var(\bar{X}) = Var(X_1/n) + \dots + Var(X_n/n) = n \cdot \sigma^2 / n^2 = \sigma^2 / n$
with $\sigma^2 = Var(X)$ and therefore its standard error is $SE_{\bar{X}} = \sqrt{\sigma^2 / n}$. As n increases, the dispersion of **estimates** around μ decreases!

In addition

- ▶ It can be shown that given n , $\bar{X}$ is the unbiased estimator with **minimum variance (the most efficient)**
- ▶ If X is normally distributed, the **distribution** of $\bar{X}$ is also **normal**; the same applies, whatever the distribution of X , if n is sufficiently high (central limit theorem).

Estimator for the population variance

Note that the variance of $\bar{X}$ depends on the variance σ^2 of the population, which is **typically unknown**. To estimate σ^2 one can use the **sample variance** S^2 :

$$S^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$

It can be shown that:

$$E(S^2) = \sigma^2$$



The sample variance S^2 is an **unbiased estimator** for σ^2 whatever the distribution of X

The so-called **unadjusted** sample variance:

$$\tilde{S}^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n}$$

is characterised by: $E(\tilde{S}^2) = \frac{n-1}{n} \sigma^2$



$\tilde{S}^2$ is a **biased estimator** for σ^2 , but it is **asymptotically unbiased**

Hands on: estimating the population mean

PEPUT 1

$x_1, x_2, \dots, x_n$ i.i.d. callr. $E[X_i] = \mu$ $Var(X_i) = \sigma^2 = 150^2$

$n =$

To assess the efficiency of a call centre, a survey is conducted on a random sample of **300** calls received in a week (dataframe `CallCenter.P*`): the day and time of each call, the reason for the contact (`Topic`), the duration (in seconds) of the wait (`TimeWaiting`) and of the conversation (`TimeTalk`) are recorded. Customers are also contacted in order to record whether they are satisfied with the service (`Satisfaction`) and whether or not their problem has been resolved (`Solved`).

The conversation duration is assumed to have a standard deviation of 150 sec.

Determine the estimate of the average conversation duration. What is the standard error of the estimator used and what is its interpretation?

```
> xbar <- mean(CallCenter.P$TimeTalk) # no missing values
> xbar
[1] 254.3733      Estimate of the population mean duration

> sigmaX <- 150 # standard deviation assumed known
> n <- nrow(CallCenter.P) # no missing values
> SE_xbar <- sigmaX/sqrt(n)
> SE_xbar
[1] 8.660254      The standard error represents the expected deviation of a generic estimate from the population mean duration
```

* This dataframe and all the other used below are in file `Lesson 13-14_Data.Rdata`

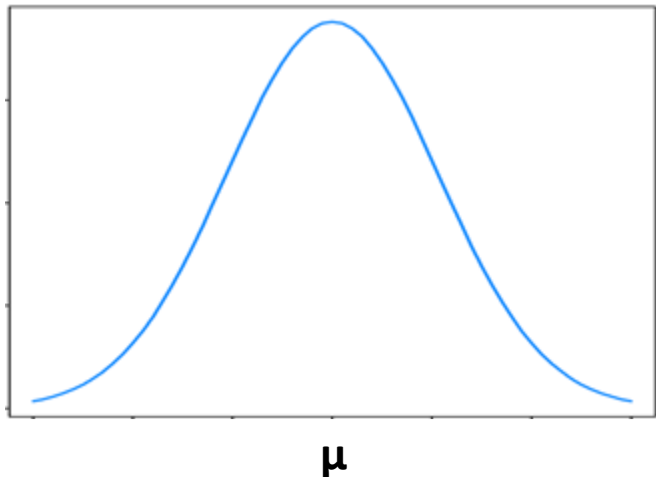
Hands on: estimating the population mean

“Since the standard error is $SE_{\bar{x}} = 8.66$ and $\bar{x} = 254.3733$ we can conclude that the expected deviation between the population mean, μ , and 254.3733 is 8.66”. Comments?

The statement is false: **8.66** is the **expected** deviation from μ of a **generic** and not of a **specific** estimate! We cannot exclude higher or lower deviations: **8.66** is only a **summary** of the deviations characterising the estimates calculated on **all** possible samples.

To further clarify this fundamental aspect, remember that since $n = 300$, $\bar{X}$ has a distribution that can be **approximated** by a normal (central limit theorem):

$$\bar{X} \sim \mathcal{N}(\mu, \sigma^2/300) = \mathcal{N}(\mu, 8.66^2)$$



The standard error measures the **dispersion** of the **estimates** obtained based on all the possible samples around μ .

We cannot rule out that the estimate obtained based on the specific available sample (254.3733) is far from μ . What we can say though is that the probability to draw a sample leading to an estimate close to μ is higher than the probability to draw a sample leading to an estimate far from μ . The lower the standard error of the estimator the higher the probability of estimates close to μ .

Hands on: estimating the population mean

How to obtain an estimator with a lower standard error? What effects on estimation?

By increasing the sample size the standard error of $\bar{X}$ decreases.

What is the sample size required to reduce the standard error to a maximum of 5 seconds?

For the standard error of the sample mean to be at most 5, we must have:

$$SE_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{150}{\sqrt{n}} \leq 5 \quad \rightarrow \sqrt{n} \geq \frac{150}{5} \quad \text{That is } \sqrt{n} \geq \frac{\sigma}{5},$$
$$\rightarrow n \geq 30^2 = 900 \quad \text{or, equivalently } n \geq \frac{\sigma^2}{5^2}$$

Assume that a sample of size 900 is drawn leading to an estimate equal to 7.24. “Since 7.24 is based upon a larger sample (900) such estimate is more reliable than the estimate, 8.66, based on a sample of size 300”. Comments?

Surely the standard error of the estimator based upon a larger sample is smaller. Therefore the concentration of the possible estimates around μ will be higher and the probability of estimates far from μ lower. Nonetheless, we cannot draw any conclusions about a **specific estimate**, because it might be that the larger sample contains some extreme or outlying cases and the smaller sample contains very “central” observations

Hands on: estimating the population mean

How would your assessment of the standard error of $\bar{X}$ and of the sample size needed to have a standard error lower than 5 change if σ^2 was not known?

It is not possible to answer precisely as in the case when the variance is known. Even so, it is possible to **estimate** σ^2 using the sample variance, S^2 ; the **estimate** of the standard error based on the available sample is: $se_{\bar{X}} = \sqrt{s^2/n}$

```
> n <- nrow(CallCenter.P)
> s2_X <- var(CallCenter.P$TimeTalk) # estimate of the variance
> s2_X
[1] 28445.97
> se_Xbar <- sqrt(s2_X/n) # estimate of the standard error
> se_Xbar
[1] 9.737552
```

Note that the estimated variance based on the sample (and the estimated standard error) is larger than the one assumed before, $\sigma^2 = 150^2 = 22500$

Based on such estimate we can approximate the sample size needed to reduce the standard error using $n \geq s^2/5^2$:

```
> s2_X/25
[1] 1137.839
```

*Based on this result, a sample sized at least **1138** should be drawn. As a precaution, it is best to consider the highest value obtained by using the highest variance for the calculations*

RStudio: a note on missing values [OPTIONAL]

For the sake of completeness, we consider the case when the dataset contains missing values, using the data in the `CallCenter.R.NoMiss` and `CallCenter.R.Miss` dataframes

```
> CallCenter.R.NoMiss$TimeWaiting
[1] 15 35 90 70 51 58 118 38 99 124 113 86
> CallCenter.R.Miss$TimeWaiting # missing value
[1] 15 35 90 70 51 58 118 38 99 124 113 86 NA
> mean(CallCenter.R.NoMiss$TimeWaiting)
[1] 74.75
> mean(CallCenter.R.Miss$TimeWaiting, na.rm=T)
[1] 74.75
> var(CallCenter.R.NoMiss$TimeWaiting)
[1] 1279.477
> var(CallCenter.R.Miss$TimeWaiting, na.rm=T)
[1] 1279.477
> # However:
> nrow(CallCenter.R.NoMiss)
[1] 12
> nrow(CallCenter.R.Miss) # the number of rows includes the missing value!
[1] 13
> complete.cases(CallCenter.R.Miss$TimeWaiting) # not NA
[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
> n.Miss<-sum(complete.cases(CallCenter.R.Miss$TimeWaiting))
> n.Miss
[1] 12
```

*When there are missing values, the number of rows in the dataframe (or the length of a data vector) **does not coincide with the number of instances!***

***Note:** We will derive the standard error of the estimators using the functions in `UBStats` for confidence intervals (introduced in the next lessons), which exclude missing values from the calculations!*

Estimator for the population proportion

Another parameter one is typically interested in is the **proportion p of cases in a population that exhibit a characteristic (coded as 'success')**, estimated on the basis of a simple random sample of n units, by measuring the r.v. $X_1, \dots, X_n$ indicating for each unit whether a success was observed (1) or not (0).

To estimate p we use the **sample proportion $\hat{P}$** .

- ▶ the r.v. $X_1, \dots, X_n$ are i.i.d. according to a Bernoulli distribution with parameter p , and $\hat{P}$ is therefore their **sample mean**. From the properties of the sample mean we have:
- ▶ $E(\hat{P}) = E(X) = p$  $\hat{P}$ is an **unbiased estimator for p** ,
- ▶ $Var(\hat{P}) = Var(X)/n = p(1 - p) / n$  The variance of $\hat{P}$ (**never known**) is smaller the larger the sample
- ▶ If n is sufficiently high, the distribution of $\hat{P}$ can be approximated by a normal distribution $\hat{P} \approx \mathcal{N}(p, p(1 - p)/n)$. Note that the variance of $\hat{P}$ is not known (but it can be estimated by substituting p with its estimate).

Hands on: estimating the population proportion PEPVS 2

The dataframe `CallCenter.P` includes, among others, a variable (`Solved`) indicating whether the customer's problem has been solved.

Estimate the proportion p of problems solved, and the standard error of the estimator.

```
> table(CallCenter.P$Solved)
 N  Y
27 273
```

Success in this case is coded as `Y`! The estimate of p is (there are no missing values)

$$\hat{p} = 273/300 = 0.91$$

$Var(\hat{P}) = p(1 - p) / n$ is not known (it depends on p) but it can be estimated by substituting p with its estimate. Thus, the estimate of the standard error is

$$se_{\hat{p}} = \sqrt{\hat{p} \cdot (1 - \hat{p}) / n} = \sqrt{0.91 \cdot (1 - 0.91) / 300} = 0.0165$$

What can you conclude about the parameter?

We **cannot** conclude that p is close to 0.91, the expected deviation from p of a generic estimate is very low, and – given the approximately normal distribution of $\hat{P}$ (high sample size, central limit theorem) – estimates are very concentrated around p and estimates far from p are unlikely, though not impossible.

Estimator for the difference between means (and proportions)

In many applications we are interested in evaluating the differences between the averages of two populations

- ▶ Cholesterol-lowering drug
 - ▶ Comparison of average cholesterol levels at the start of the treatment and after a certain period of treatment
 - ▶ Comparison of average cholesterol levels in patients treated with two different drugs
- ▶ Gender gap: comparison of average salary levels for men and women in the same position
- ▶ Marketing campaigns
 - ▶ Average expenditure before and after a sales campaign
 - ▶ Comparison of the percentage of customers clicking on two different banners
- ▶ Comparison of the percentage of users opening the link to an ad posted on different social media (e.g. facebook, instagram)

Estimators for the difference between means

Problem setting:

Population 1 - described by the r.v. X	Population 2 - described by the r.v. Y
$X_1, \dots, X_{n_X}$ i.i.d. sample of size n_X from X with mean μ_X and variance σ_X^2	$Y_1, \dots, Y_{n_Y}$ i.i.d. sample of size n_Y from Y with mean μ_Y and variance σ_Y^2
$\bar{X} = (X_1 + \dots + X_{n_X})/n_X$	$\bar{Y} = (Y_1 + \dots + Y_{n_Y})/n_Y$
$s_X^2 = \sum_{i=1}^{n_X} \frac{(X_i - \bar{X})^2}{n_X - 1}$	$s_Y^2 = \sum_{i=1}^{n_Y} \frac{(Y_i - \bar{Y})^2}{n_Y - 1}$

The parameter of interest is the difference between the means of the two populations $\mu_X - \mu_Y$

Estimators for the difference between means

To estimate the difference between the means of two populations described by two r.v. X and Y on the basis of two samples $X_1, \dots, X_{n_X}$ and $Y_1, \dots, Y_{n_Y}$, it is natural to use the estimator: $\bar{X} - \bar{Y}$. **Whatever the distribution of X and Y we have:**

▶ $E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_X - \mu_Y$

$(\bar{X} - \bar{Y})$ is therefore an **unbiased estimator for $\mu_X - \mu_Y$**

- ▶ The variance, and thus the standard error, of the estimator **depends on the relationships between the two populations (and thus between the samples drawn from them) and on the assumptions on the joint distribution of X and Y !**
- ▶ The **distribution** of $\bar{X} - \bar{Y}$ also depends on the relationship between X and Y and on their joint distribution. In general, with **large** samples the distribution of the sample means can be approximated with a **normal** (even if the variances are not known). However, some clarifications are necessary - especially for **small** samples; we will discuss about this when knowing the distributions of the estimators will be of crucial importance.

Estimators for the difference between means

We deal with two specific cases:

- ▶ **Independent** samples. We are interested in comparing the means of two populations on the basis of samples, **possibly of different sizes**, drawn **independently** (e.g. average wages for men and women, profitability of customers reached by different marketing campaigns, sales volumes in restructured and non-restructured stores,...). This is the case where the two samples are made up of different statistical units, so that it makes sense to assume that the results observed on one sample are in no way linked to the results observed on the other.
- ▶ **Paired** samples. In this case, the two samples represent two measurements from the **same statistical units**. For example, the same variable is measured at different times, or under different conditions (e.g. the average amount spent by a customer before and after a promotional campaign, or at different times of the year). Thus, the two samples have the **same size** and are **paired** in the sense that two measurements are available for the same unit.

Estimator for the difference between means: independent samples

In the case of **independent** samples, the sample averages $\bar{X}$ and $\bar{Y}$ are **independent** and thus **uncorrelated**, and the **variance** of the estimator $(\bar{X} - \bar{Y})$ is:

$$Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y} \quad \Rightarrow \quad SE(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

The case where both variances are known is very rare.

- ▶ If the variances **are not known** and are assumed to be **different**, $Var(\bar{X} - \bar{Y})$ is estimated by replacing σ_X^2 and σ_Y^2 with the sample variances obtained from the two samples, S_X^2 and S_Y^2 .
- ▶ If the two **unknown** variances are **assumed equal**, so that $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, the common variance is estimated using the so-called **pooled sample variance**:

$$S_{Pool}^2 = \frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_X + n_Y - 2} \quad \text{weighted average of sample variances}$$

σ_X^2 and σ_Y^2 are both replaced with S_{Pool}^2 to estimate variance and standard error of $(\bar{X} - \bar{Y})$.

Estimators for the difference between means: paired samples

In the case of **paired samples** of size n , two measurements are available for each sample unit: X_i and Y_i ; the differences $D_i = X_i - Y_i$ can therefore be considered as n measurements from the r.v. $D = X - Y$.

One can then refer to the sample mean of the differences, $\bar{D} = \bar{X} - \bar{Y}$.

- ▶ Clearly $E(\bar{D}) = \mu_D = E(\bar{X} - \bar{Y}) = \mu_X - \mu_Y$,
- ▶ As for the variance of $\bar{D}$, by the properties of the sample mean we have that:

$$Var(\bar{D}) = Var(D)/n = \sigma_D^2/n$$

Using the properties of linear combinations of r.v. we have that

$$Var(D) = Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}$$

In this case, it is very rare to assume that the variance of $D = (X - Y)$ is known, as it would be necessary to make assumptions directly on the variance of differences, σ_D^2 , **or** on the variances of X and Y **and** on their covariance (or correlation)

Estimators for the difference between means: paired samples

To **estimate** the variance of $\bar{D} = \bar{X} - \bar{Y}$ it is possible to proceed in the same way as for the sample mean, and use the **sample variance** calculated on the differences:

$$s_D^2 = \sum_{i=1}^n \frac{(D_i - \bar{D})^2}{n - 1}$$

which is clearly related to the variances and sample covariance of the two samples:

$$s_D^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n - 1} + \sum_{i=1}^n \frac{(Y_i - \bar{Y})^2}{n - 1} - 2 \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1} = s_X^2 + s_Y^2 - 2s_{XY}$$

σ_D^2 is **replaced** with s_D^2 to estimate variance and standard error of $(\bar{X} - \bar{Y})$.

Hands on: estimating the difference between means PEPUS 3

The dataframe **Insurance** (in `Lesson 13_Data.Rdata`) contains information on reimbursements (**refunds**, in \$) for healthcare expenses to (unrelated) subscribers with different socio-demographic characteristics (**age**, **sex**, **bmi** - body mass index - **children**, **smoker**, **region**).

Estimate the difference in the average amount of expenditure reimbursed to smokers and non-smokers, limiting attention to those who do not have children.

The two samples are **independent** as the subscribers have no links to each other

```
> # extraction of the two samples
> refund.smoke <- Insurance$refund[Insurance$smoker=="yes" &
+ Insurance$children==0]
> refund.nosmoke <- Insurance$refund[Insurance$smoker=="no" &
+ Insurance$children==0]
> mean_x <- mean(refund.smoke)
> mean_y <- mean(refund.nosmoke)
> mean_x_y <- mean_x-mean_y

> mean_x_y
[1] 23729.57
```

The average amount reimbursed to smokers is higher. Obviously, there could be other factors that influence spending and that could be 'controlled for', but overall there is a difference between the two means

Hands on: estimating the difference between means

The dataframe `Insurance` contains information on reimbursements (`refunds`, in \$) for healthcare expenses to subscribers (unrelated) with different socio-demographic characteristics (`age`, `sex`, `bmi` - body mass index - `children`, `smoker`, `region`).

Estimate the standard error of the estimator of the difference in the average amount of expenditure reimbursed to smokers and non-smokers, limiting attention to those who do not have children.

```
> nx<-length(refund.smoke) # no missing values!  
> ny<-length(refund.nosmoke)  
> var_x<-var(refund.smoke)  
> var_y<-var(refund.nosmoke)  
> # S2_pooled  
> var_pooled<- ((nx-1)*var_x+(ny-1)*var_y) / (nx+ny-2)  
> var_pooled  
[1] 54284441
```

$$\longrightarrow s_{Pool}^2 = \frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}$$

```
> se_x_y<-sqrt(var_pooled*((1/nx)+(1/ny)))  
> se_x_y  
[1] 768.3133
```

$$\longrightarrow se_{\bar{X}-\bar{Y}} = \sqrt{\frac{s_{Pool}^2}{n_X} + \frac{s_{Pool}^2}{n_Y}}$$

Hands on: estimating the difference between means

PEPVS 4

A clinical study conducted on 44 patients with high (bad) cholesterol levels to evaluate the efficacy of a cholesterol-lowering drug reported the following results:

Cholesterol level at start of treatment: $\bar{x}_{ini} = 155$; $s_{ini} = 19.9$

Cholesterol level at 8 weeks: $\bar{x}_{8w} = 122$; $s_{8w} = 24.5$; $s_{ini,8w} = 23.4$

SAMPLE COVARIANCE

Estimate the average reduction in cholesterol level following treatment. Standard error?

The two samples are **paired**: they relate to two measurements on the same individual

We obtain the estimate of the difference between means and the estimate of the std error from the available sample statistics. If we had the data for each patient, we could determine them from the differences between the data on cholesterol levels before and after treatment for each patient!

Estimate of the difference between means:

$$\bar{x}_{ini} - \bar{x}_{8w} = 155 - 122 = 33$$

Estimate of the variance of the difference, $D = X_{ini} - X_{8w}$

$$s_D^2 = s_{ini}^2 + s_{8w}^2 - 2s_{ini,8w} = 19.9^2 + 24.5^2 - 2 \cdot 23.4 = 949.46$$

Estimate of the standard error of the difference between sample means, $\bar{D} = \bar{X}_{ini} - \bar{X}_{8w}$

$$se_{\bar{X}_{ini} - \bar{X}_{8w}} = \sqrt{s_D^2/n} = \sqrt{949.46/44} = 4.645281$$

The expected deviation of a generic estimate of the difference between means is about 4.65.

dependent
samples

Hands on: estimate and estimator of difference between means

The dataframe `Fatalities_US` contains data on the number of accidents (per 100,000 inhabitants) registered in 48 US states (`Accidents`). General information on the states is available (unemployment rate, % of drivers aged between 15 and 24, `YoungDrivers`), besides the number of accidents at night (`Night`) and due to alcohol (`Alcohol`). In order to fight alcohol-related accidents, an awareness-raising campaign was carried out, following which a new survey of the number of such accidents (per 100,000 inhabitants, `AlcoholPostCampaign`) was collected for each state.

Assess the drop in the average number of alcohol-related accidents after the campaign.

The two samples are **paired**: they refer to observations of the same state before and after the campaign.

```
> diff <- Fatalities_US$Alcohol-Fatalities_US$AlcoholPostCampaign
> mean_d <- mean(diff)
> se_diff <- sd(diff)/sqrt(length(diff)) # estimate of standard error
> mean_d
[1] 1.281191
> se_diff
[1] 0.3328564
```

Check on the R script that the same result is obtained from the aggregated statistics

Estimators for the difference between proportions

Problem setting: we are interested in comparing the proportions of a certain phenomenon in two different populations

Population 1	Population 2
$X_1, \dots, X_{n_X}$ i.i.d sample of size n_X from X distributed according to a Bernoulli with parameter p_X (i.e. X takes 1 or 0 depending on whether a success is observed or not, and p_X = proportion of successes in the population), with $E(X) = p_X$ and $Var(X) = p_X(1 - p_X)$	$Y_1, \dots, Y_{n_Y}$ i.i.d sample of size n_Y from Y distributed according to a Bernoulli with parameter p_Y (i.e. Y takes 1 or 0 depending on whether a success is observed or not, and p_Y = proportion of successes in the population), with $E(Y) = p_Y$ and $Var(Y) = p_Y(1 - p_Y)$
$\hat{P}_X = (X_1 + \dots + X_{n_X})/n_X$ = sample proportion of successes	$\hat{P}_Y = (Y_1 + \dots + Y_{n_Y})/n_Y$ = sample proportion of successes

The parameter of interest is the difference between the proportions in the two populations:
 $p_X - p_Y$

Estimators for the difference between proportions

To estimate the difference between the proportions of 'successes' in two populations, it is natural to use the estimator $\hat{P}_X - \hat{P}_Y$ where $\hat{P}_X$ and $\hat{P}_Y$ are the sample means of two samples whose units have Bernoulli distributions with parameters p_X and p_Y respectively.

For the properties of sample means:

► $E(\hat{P}_X - \hat{P}_Y) = E(\hat{P}_X) - E(\hat{P}_Y) = p_X - p_Y$

$(\hat{P}_X - \hat{P}_Y)$ is therefore an **unbiased estimator** for $p_X - p_Y$

- The variance of the estimator **depends on the relationships between X and Y** . In the case of **independent** samples:

$$\begin{aligned} \text{Var}(\hat{P}_X - \hat{P}_Y) &= \text{Var}(\hat{P}_X) + \text{Var}(\hat{P}_Y) = \frac{\text{Var}(X)}{n_X} + \frac{\text{Var}(Y)}{n_Y} \\ &= \frac{p_X(1 - p_X)}{n_X} + \frac{p_Y(1 - p_Y)}{n_Y} \end{aligned}$$

$\text{Var}(\hat{P}_X - \hat{P}_Y)$ is unknown (it depends on p_X and p_Y), but can be estimated by replacing p_X and p_Y with their estimates, i.e. the observed sample proportions.

Hands on: estimating the difference between proportions

In order to determine which of two banners, A or B, is more effective, users of a social network are shown one of the two banners chosen at random. We are interested in the Click-Through Rate (CTR) of the two banners, i.e. the ratio between the number of clicks on each banner and the number of banner views served by the ad server (impressions). Banner A is viewed by 2364 users and generates 456 clicks. Banner B is viewed by 2323 users and generates 611 clicks.

Estimation of the difference between the CTRs? Standard error of the estimator?

The CTR is the sample proportion of clicks on impressions; the standard error of the estimator can be **estimated** as the two experiments and thus the two samples are independent.

Estimates of the two proportions and of their difference:

$$\hat{p}_A = 456/2364 = 0.1929; \hat{p}_B = 611/2323 = 0.263; \hat{p}_A - \hat{p}_B = 0.1929 - 0.263 = -0.0701$$

Estimates of the variances of the sample proportions, $\hat{P}_A$ e $\hat{P}_B$,

$$\widehat{Var}(\hat{P}_A) = 0.1929 \cdot (1 - 0.1929)/2364 = 0.000066$$

$$\widehat{Var}(\hat{P}_B) = 0.263 \cdot (1 - 0.263)/2323 = 0.000083$$

Estimate of the standard error of the difference between the two sample proportions, $\hat{P}_A$ e $\hat{P}_B$,

$$se_{\hat{P}_A - \hat{P}_B} = \sqrt{\widehat{Var}(\hat{P}_A) + \widehat{Var}(\hat{P}_B)} = \sqrt{0.000066 + 0.000083} = 0.0122$$

Recap

Parameter	Estimator	Estimate	Samples	Standard error, SE	Standard error estimate, se
μ	$\bar{X}$	$\bar{x}$		$\sigma/\sqrt{n}$	$s/\sqrt{n}$
p	$\hat{P}$	$\hat{p}$		$\sqrt{p(1-p)/n}$	$\sqrt{\hat{p}(1-\hat{p})/n}$
$\mu_X - \mu_Y$	$\bar{X} - \bar{Y}$	$\bar{x} - \bar{y}$	Independent	$\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$	$\sigma_X^2 = \sigma_Y^2 \rightarrow \sqrt{\frac{s_{Pool}^2}{n_X} + \frac{s_{Pool}^2}{n_Y}}$
					$\sigma_X^2 \neq \sigma_Y^2 \rightarrow \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$
			Paired	$\sqrt{\frac{\sigma_D^2}{n}} = \sqrt{\frac{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}{n}}$	$\sqrt{\frac{s_D^2}{n}} = \sqrt{\frac{s_X^2 + s_Y^2 - 2s_{XY}}{n}}$
$p_X - p_Y$	$\hat{P}_X - \hat{P}_Y$	$\hat{p}_X - \hat{p}_Y$	Independent	$\sqrt{\frac{p_X(1-p_X)}{n_X} + \frac{p_Y(1-p_Y)}{n_Y}}$	$\sqrt{\frac{\hat{p}_X(1-\hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1-\hat{p}_Y)}{n_Y}}$

From point estimate to interval estimate [READING]

Point estimation: sample data is used to determine a **single value** to estimate an unknown population parameter.

In general, a point estimator coincides with the parameter of interest with zero probability! It is virtually impossible for a sample to provide an estimate that coincides with the parameter and – even if it did – this would not be known. With reference to a point estimator, it is always **necessary to provide information on its standard error in order to correctly convey the dispersion of estimates around the parameter.**

We may be interested in proposing estimators that **'incorporate' information about the dispersion around the parameter**, as well as in somehow **controlling the probability that the estimator 'covers' the parameter of interest.**

Interval estimation: sample data are used to determine an **interval estimator** (whose extremes are functions of the sample observations) that contains an unknown population parameter with **a certain probability**